Principal Differential Analysis

Pang Du

Department of Statistics
Virginia Tech

Define the problem

- Recall that an order-m linear differential operator is defined as

$$L = \beta_0 I + \ldots + \beta_{m-1} D^{m-1} + D^m,$$

where $\beta_0, \ldots, \beta_{m-1}$ are coefficient functions.

- Goal: Given functional profiles $x_i, i = 1, \ldots, N$, we want to find an L that **comes as close as possible** to satisfying the homogeneous linear differential equation

$$Lx_i = 0, i = 1, \ldots, N.$$

The least squares criterion:

$$\mathrm{SSE}_{\mathrm{PDA}}(L|\mathbf{x}) = \sum_{i=1}^{N} \int [Lx_i(t)]^2 dt = \sum_{i=1}^{N} \|Lx_i\|^2.$$

The problem with forcing functions

- Suppose there are forcing functions $f_i, i = 1, \ldots, N$.
- Goal: Given functional profiles x_i and forcing functions f_i, $i = 1, \ldots, N$, we want to find an L that comes as close as possible to satisfying the **nonhomogeneous** linear differential equation

$$Lx_i = f_i, i = 1, \ldots, N.$$

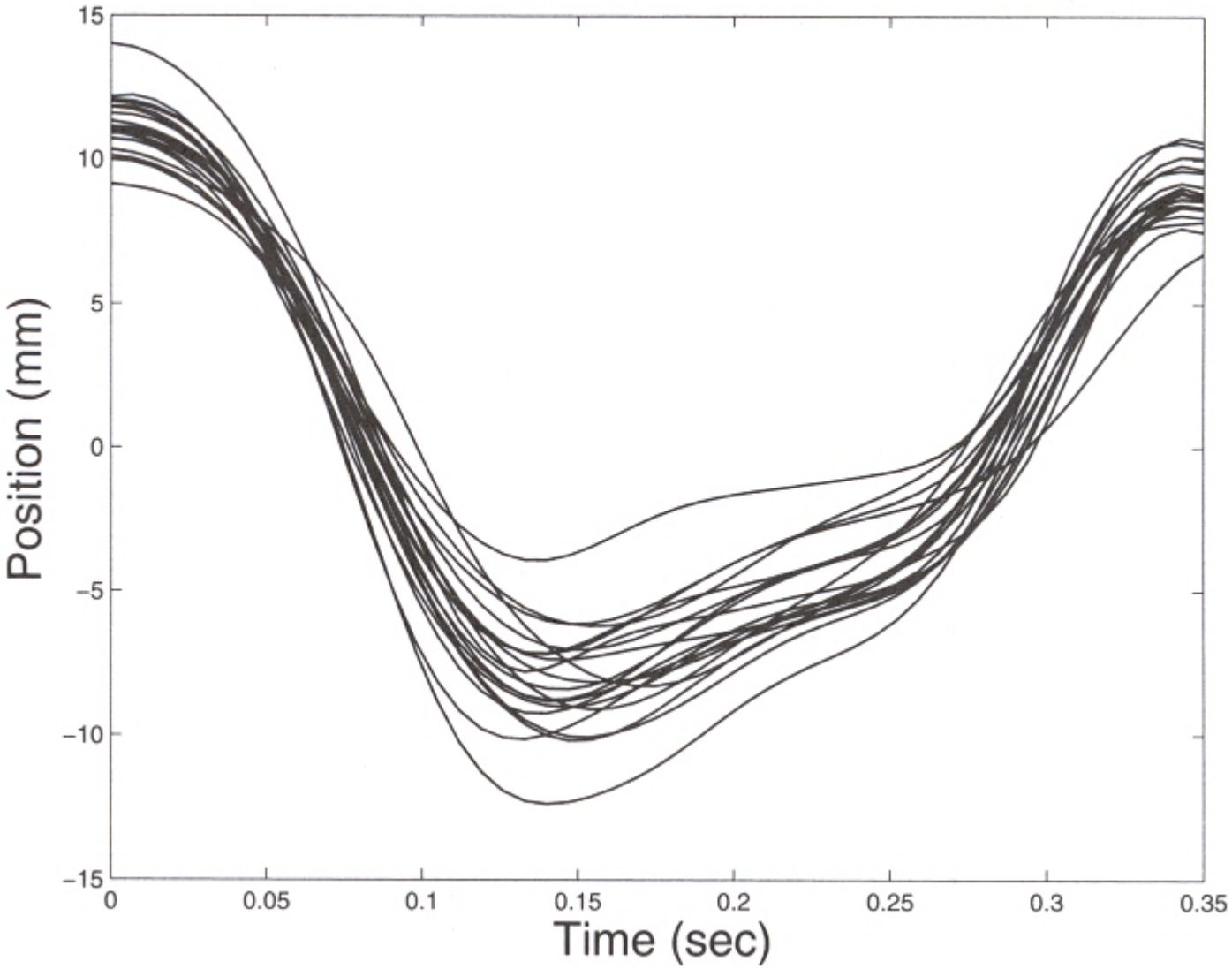
The least squares criterion:

$$\mathrm{SSE}_{\mathrm{PDA}}(L|\mathbf{x}, \mathbf{f}) = \sum_{i=1}^{N} \int [Lx_i(t) - f_i(t)]^2 dt = \sum_{i=1}^{N} \|Lx_i - f_i\|^2.$$

The reference and background

- Reference: Ramsay, Munhall, Gracco and Ostry (1996). "Functional data analyses of lip motion". *Journal of the Acoustical Society of America* **99**: 3718-3727.
- A subject was required to say the syllable "bob" $N = 20$ times.
- Plots show 20 records of position of the center of the lower lip.
- Preprocessing includes smoothing and use of functional PCA to identify the direction in which most of the motion was found.

Data on lip movements



Some biomechanics of lip movement

- Lip movement is a type of mechanic systems that is generally described by a 2nd-order linear differential equation:

$$Lx = \beta_0 x + \beta_1 Dx + D^2 x = 0.$$

- β_0: reflects the position-dependent force applied to the system at position x.
- β_1: indicates influences on the system that are proportional to velocity, often internal or external frictional forces or viscosity in mechanical systems.
- $\beta_0 > 0, \beta_1 = 0$: a system with sinusoidal or harmonic motion, with period $2\pi/\sqrt{\beta_0}$; β_0 is called the **spring constant**.
- $\beta_0 = \beta_1 = 0$: the system is in linear motion.

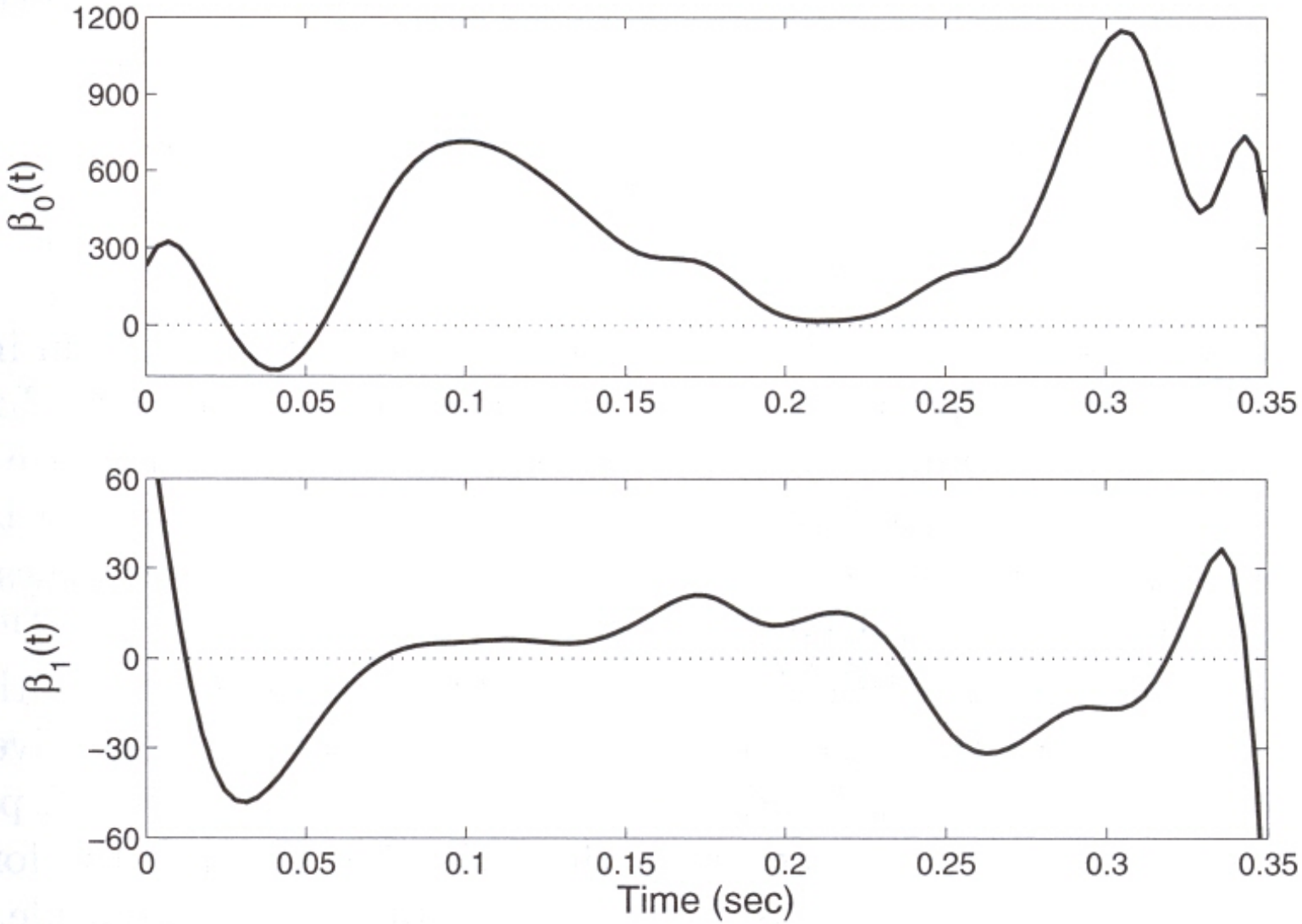
Some biomechanics: discriminant

Discriminant of the previous mechanic system is defined as $d = (\beta_1/2)^2 - \beta_0$.

- If $\beta_1 > 0$,
- $d < 0$: an under-damped system tending to exhibit oscillation that gradually disappears.
- $d > 0$: an over-damped system that becomes stable so quickly that no oscillation is observed.
- $d = 0$: a critically damped system that exhibits non-oscillatory motion that decays rapidly to zero.
- $\beta_1 < 0$ represents systems that run out of control with/without oscillation.

With time-varying coefficients, these parameters simply represents the instantaneous state of the system.

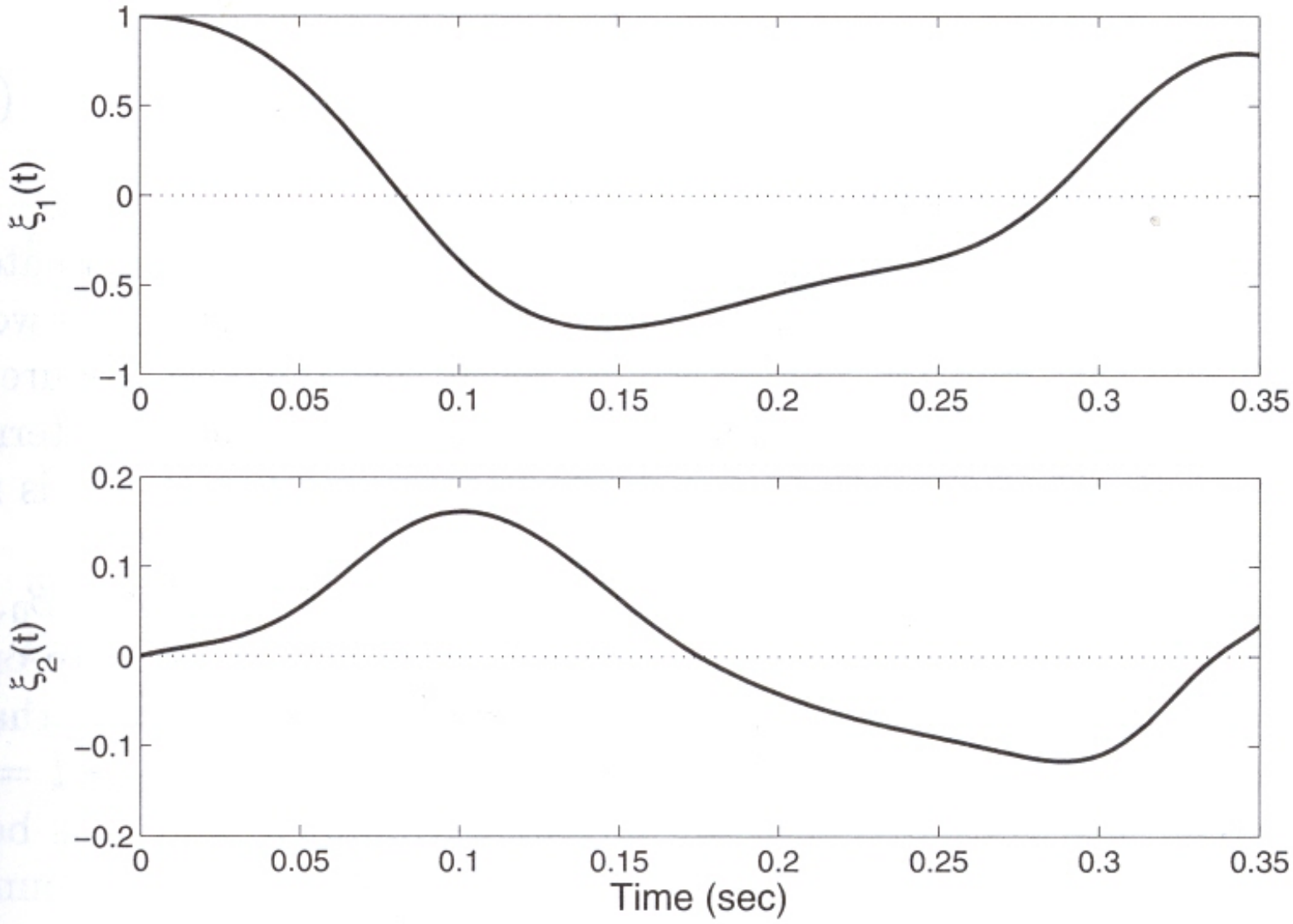
Lip movement: estimated $\beta_j(t)$



What do we see?

- $\beta_0(t) > 0$ on the whole domain except for a brief episode near the beginning.
- $\beta_1(t) \approx 0$ in the middle of the time period corresponding to the near-linear (static) phase of lip movement.

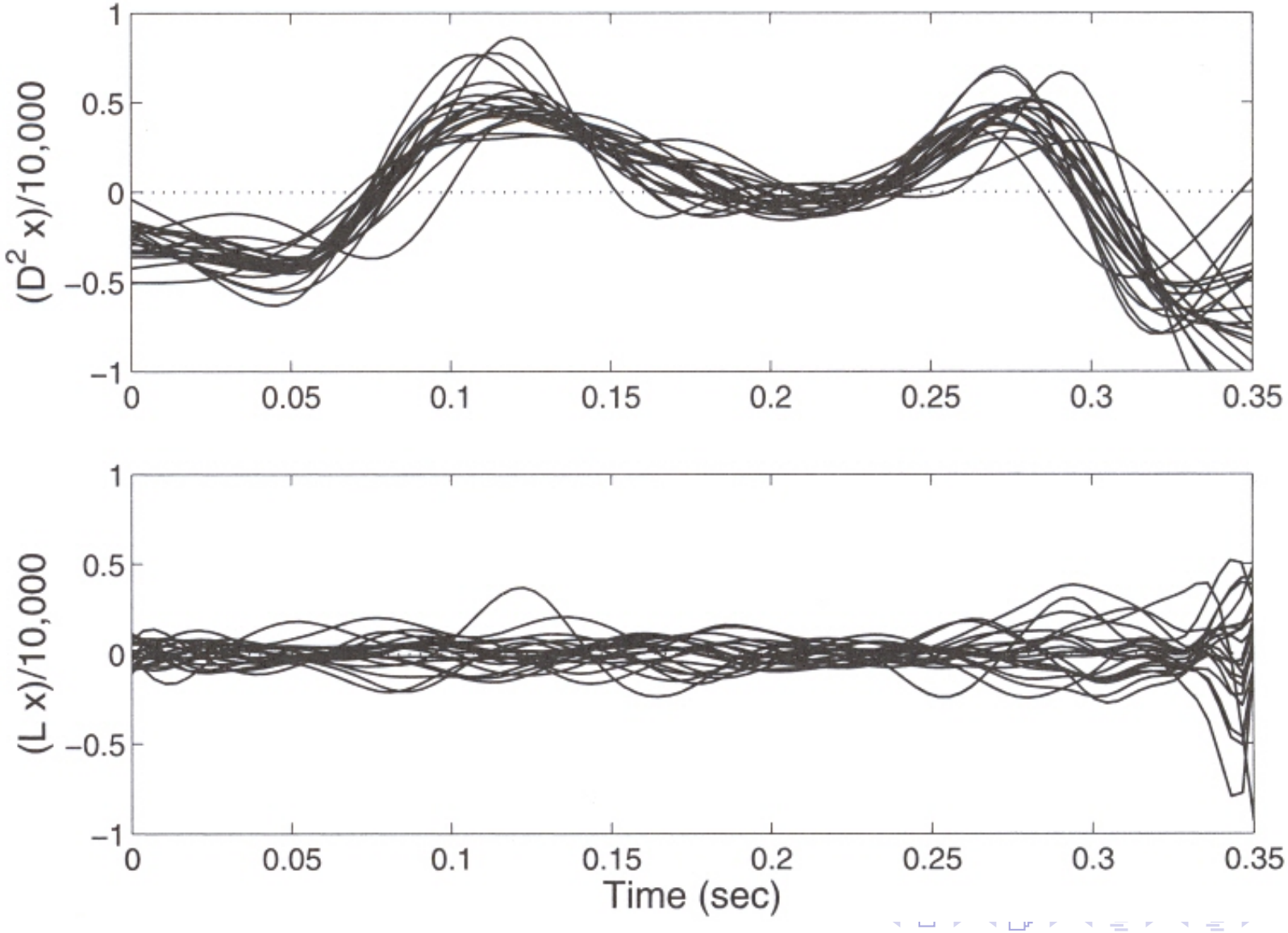
Lip movement: Solutions to $Lx = 0$



Visualizing the PDA

To look at how effective the differential operator L is at annihilating variation in the x_i, we can plot the *empirical forcing functions* Lx_i.

Lip movement: Empirical forcing



A version of R-square

- Recall that

$$\mathrm{SSE}_{\mathrm{PDA}} = \sum_{i=1}^{N} \int [\hat{L}x_i(t)]^2 dt = \sum_{i=1}^{N} \|\hat{L}x_i\|^2.$$

- Define

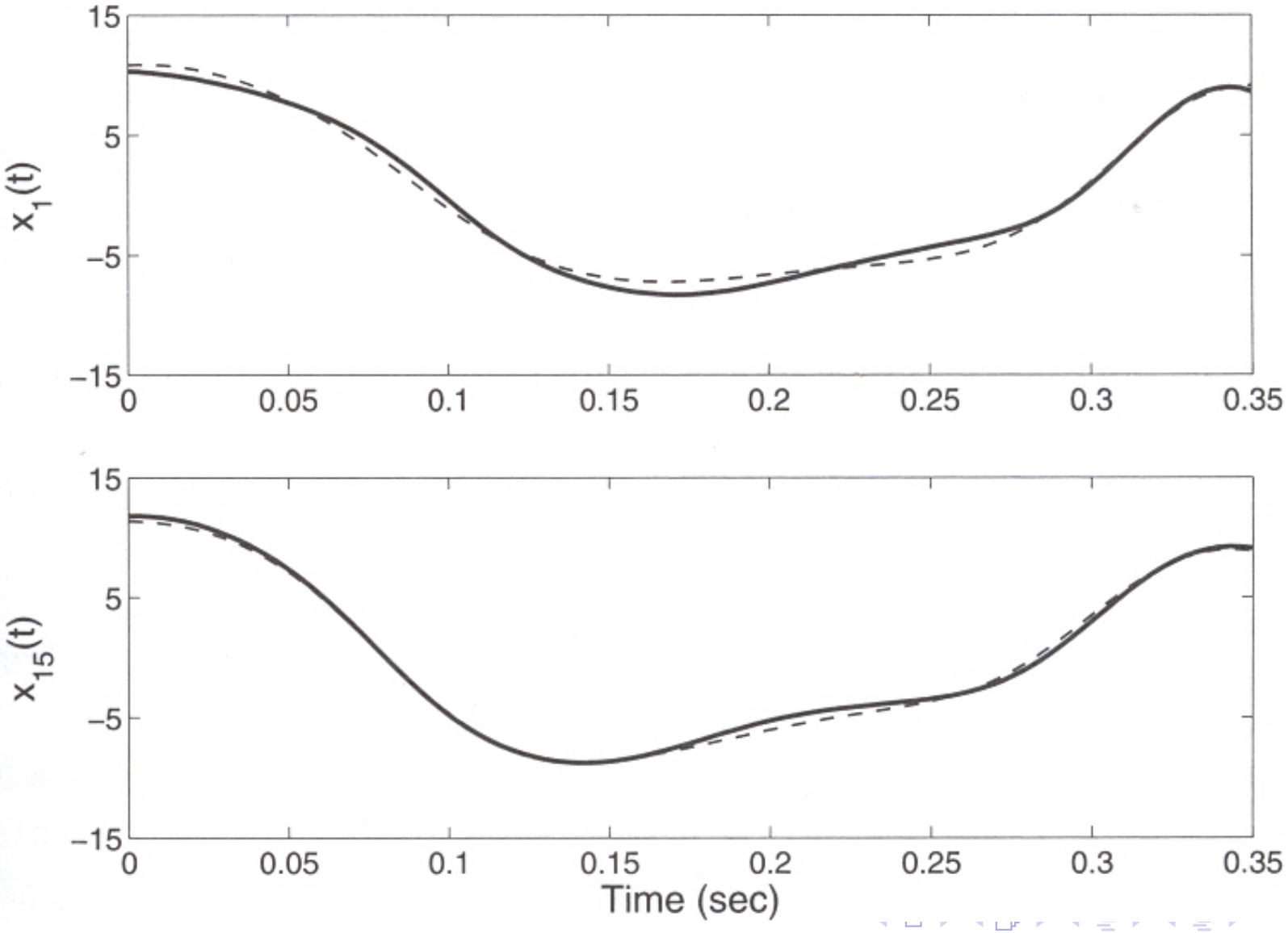
$$\mathrm{SSY}_{\mathrm{PDA}} = \sum_{i=1}^{N} \int [D^2 x_i(t)]^2 dt = \sum_{i=1}^{N} \|D^2 x_i\|^2.$$

- Define the R-square:

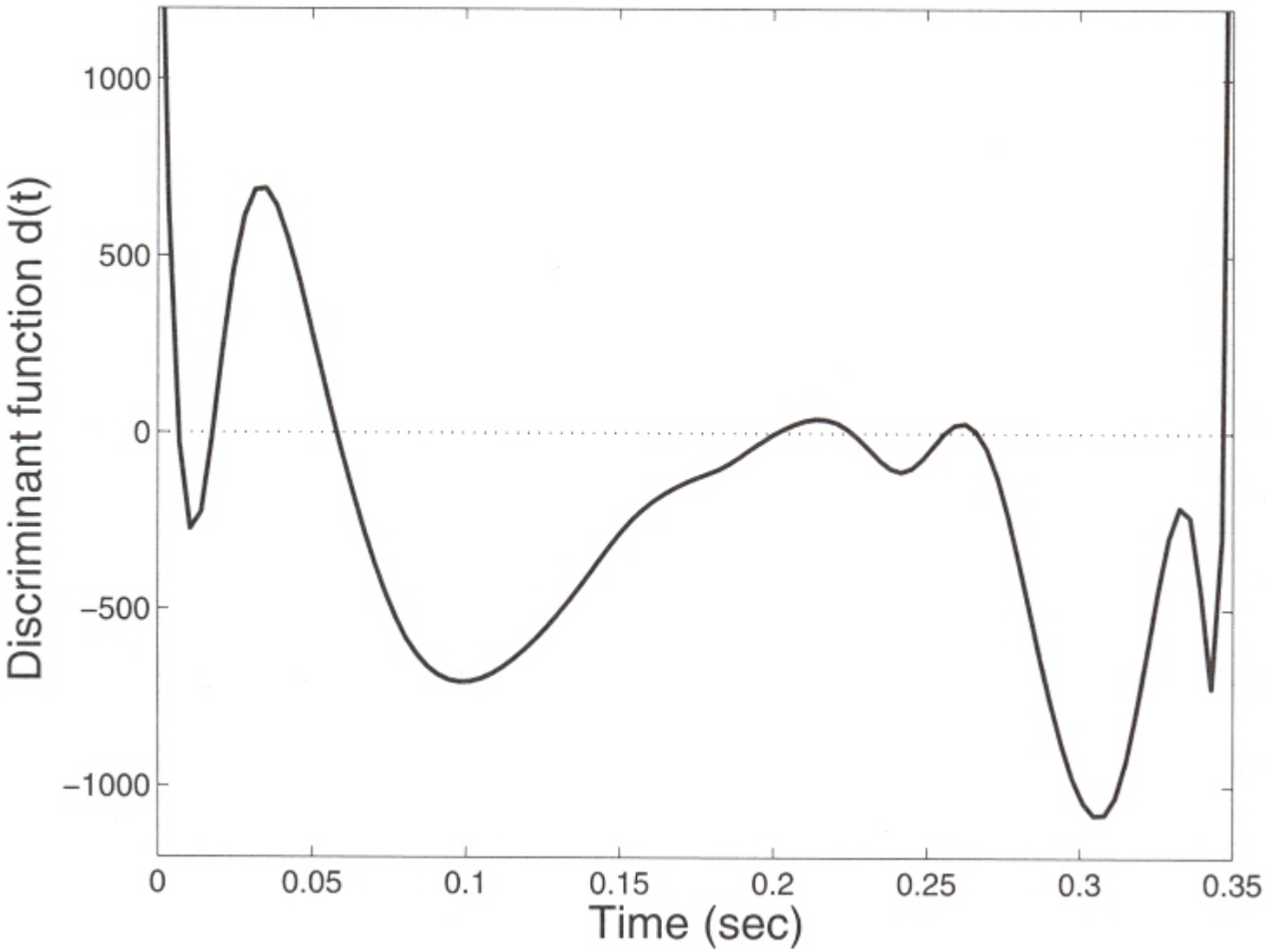
$$\mathrm{RSQ}_{\mathrm{PDA}} = (\mathrm{SSY}_{\mathrm{PDA}} - \mathrm{SSE}_{\mathrm{PDA}})/\mathrm{SSY}_{\mathrm{PDA}}.$$

- Lip example: $\mathrm{SSY}_{\mathrm{PDA}} = 90.4 \times 10^6$ and $\mathrm{SSE}_{\mathrm{PDA}} = 7.7 \times 10^6$, yielding an $\mathrm{RSQ}_{\mathrm{PDA}} = 0.92$.

Two lip position functions



Lip movement: the discriminant



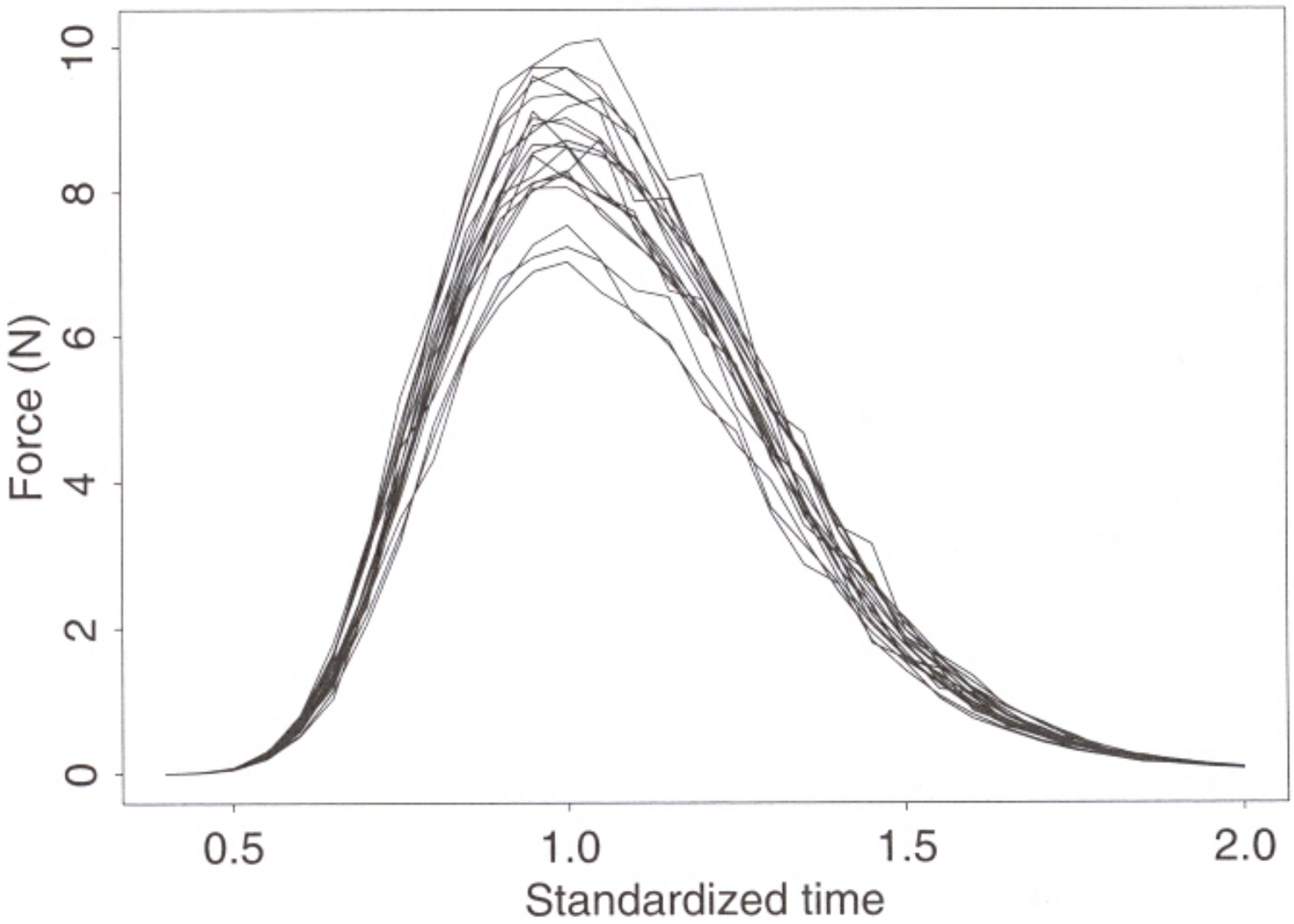
Interpret the discriminant function

- $d \approx 0$ over the interval $0.18 \leq t \leq 0.26$: a critically-damped system whose behavior may be under external control.
- $d < 0$ around $t = 0.08, 0.30$: an under-damped system whose local behavior is like a gradually slowing down bouncing spring.
- The external input to lip motions tends to be concentrated in the brief period near $t = 0.20$, when the natural tendency for the lip to close is retarded in order to allow for the articulation of the vowel.

The reference and background

- Reference: Ramsay, Wang and Flanagan (1995). "A functional data analysis of the pinch force of human fingers". *Applied Statistics* **44**: 17-30.
- The data consisted of $N = 20$ records of the force exerted on a meter during a brief pinch by the thumb and forefinger.
- The subject was required to maintain a certain background force on a force meter and then to squeeze the meter aiming at a specified maximum value, returning afterwards to the background level.
- The purpose of the biomechanical experiment was to study the neurophysiology of the thumb-forefinger muscle group.

Pinch force data



A theoretical model

- Some theoretical considerations suggest that the model
$$y_i(t) = C_i \exp[-\log^2 t/(2\sigma^2)]$$
offers a good account of any specific force function.
- Preprocessing conformed to a fixed shape parameter $\sigma^2 = 0.05$.
- The annihilating differential operator for the theoretical model
$$L_0 = [(t\sigma^2)^{-1} \log t]I + D.$$

The goals of our PDA

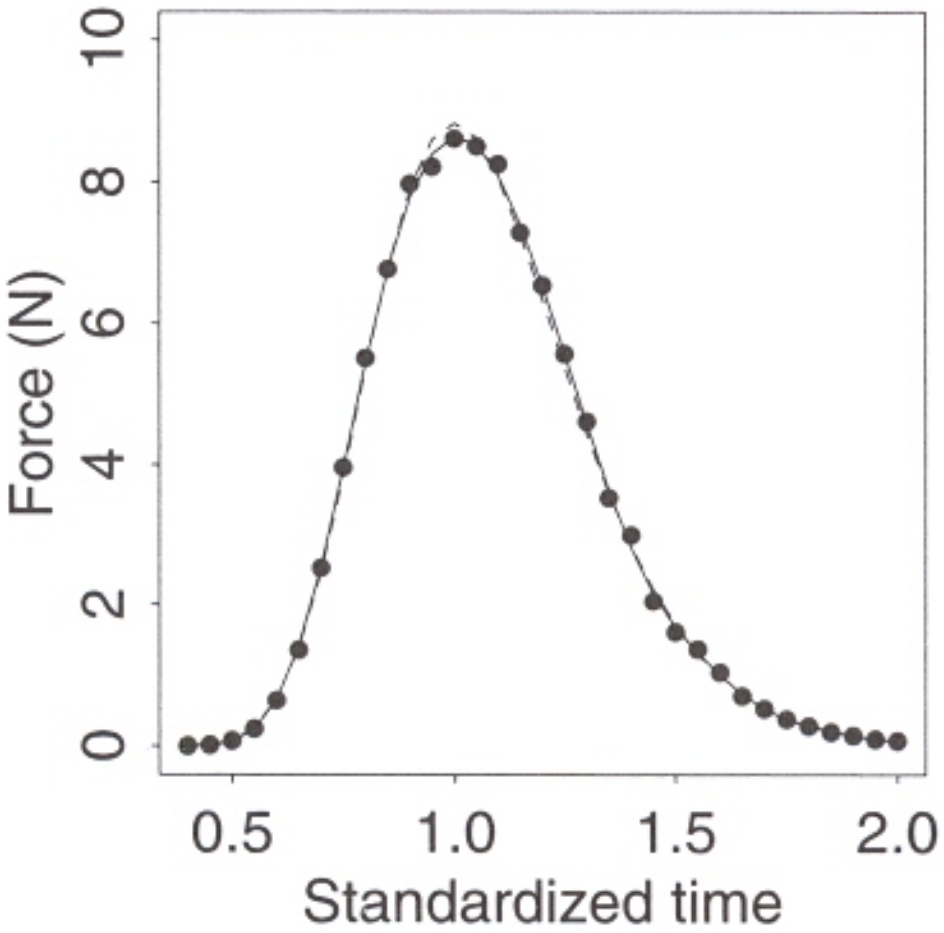
- Compare the theoretical operator with an estimated first order differential operator $L = \beta_0(t)I + D$.
- Compare the theoretical weight function $\omega_0(t) = (t\sigma^2)^{-1} \log t$ with the estimated $\beta_0(t)$.

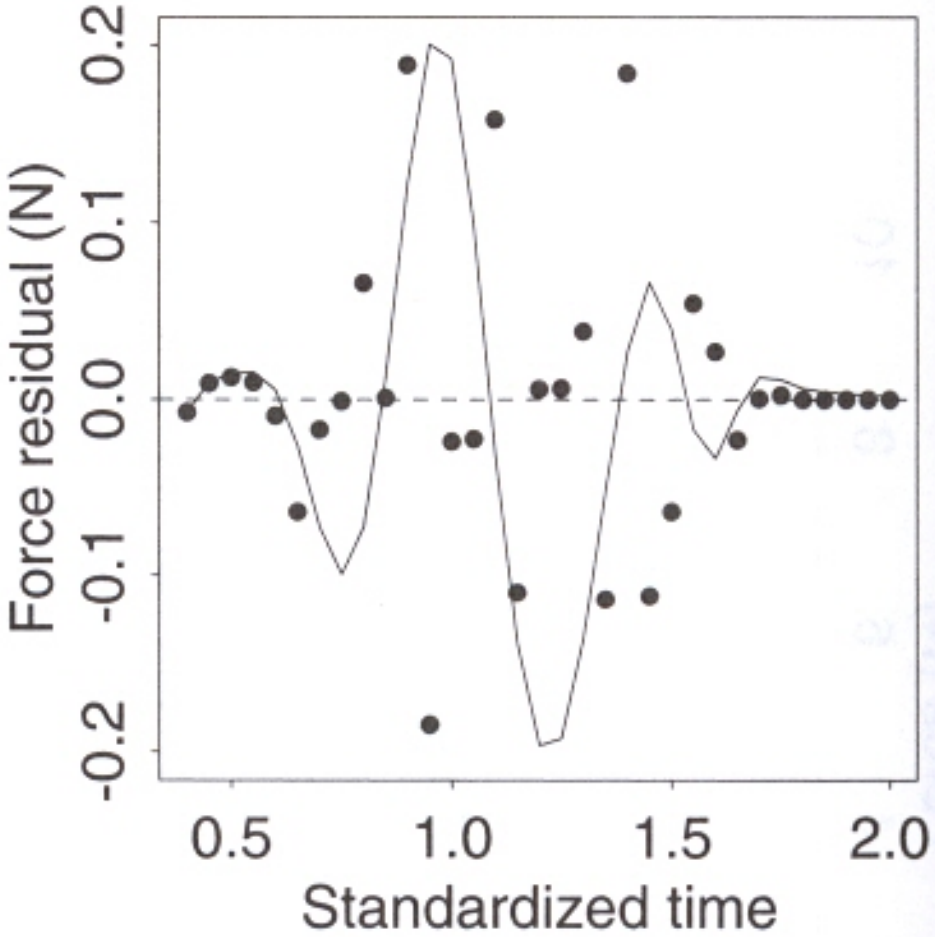
Smoothing the data

- A two-step smoothing was used.
- The data shown were first smoothed by splines with a third-derivative penalty.
- The plot showed nonconstant variation over time.
- Introducing weights σ_j^2: smoothed the logs of the standard deviations of the residuals from the first smoothing step, and take the squares of the exponentially-transformed smooth values as the weights σ_j^2.
- Re-smooth by the PWLS criterion

$$\text{PENSSE}_\lambda(\mathbf{x}|\mathbf{y}) = \sum_j [y_j - x(t_j)]^2/\sigma_j^2 + \lambda\|D^3 x\|^2.$$

Record one: smoothing (solid) vs theory (dashed)





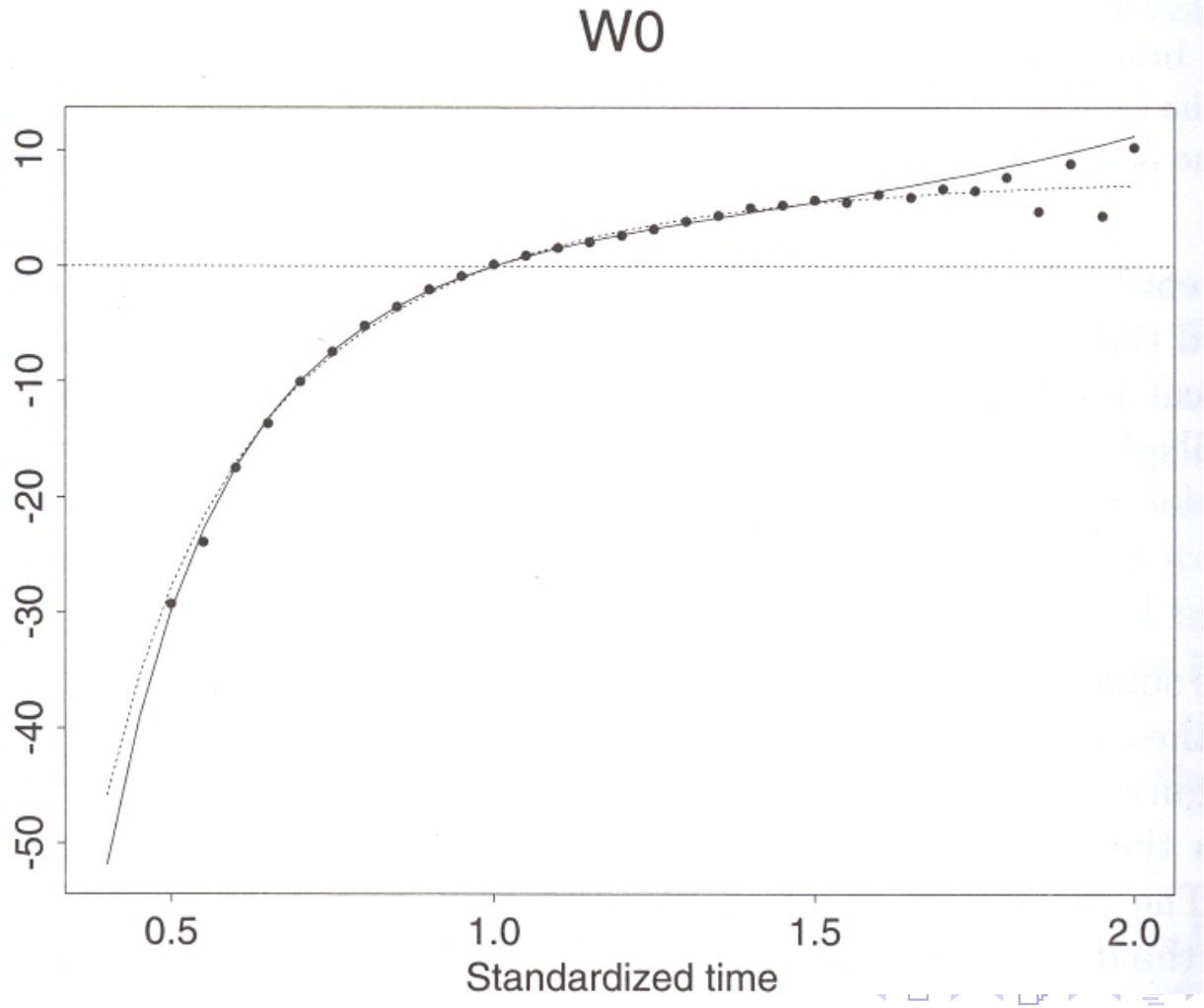
Improving over the theoretical model?

- Applied both the point-wise and basis expansion procedures for estimating β_0.
- The basis functions for the basis expansion procedure were

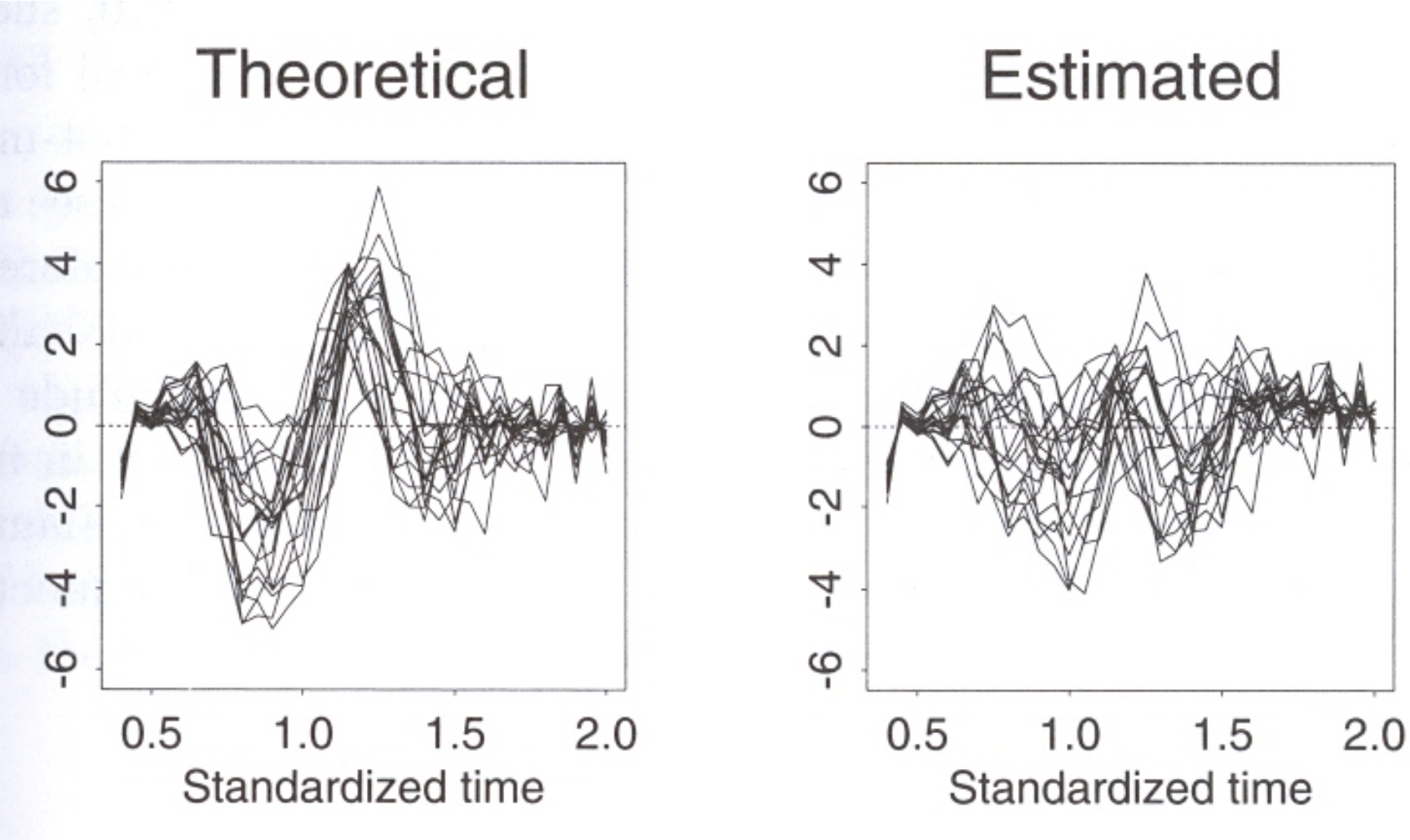
$$\phi(t) = (t^{-1}\log t, 1, t-1, (t-1)^2)'.$$

Weight function estimates

Points: point-wise. Solid: basis expansion. Dotted: theoretical weight.

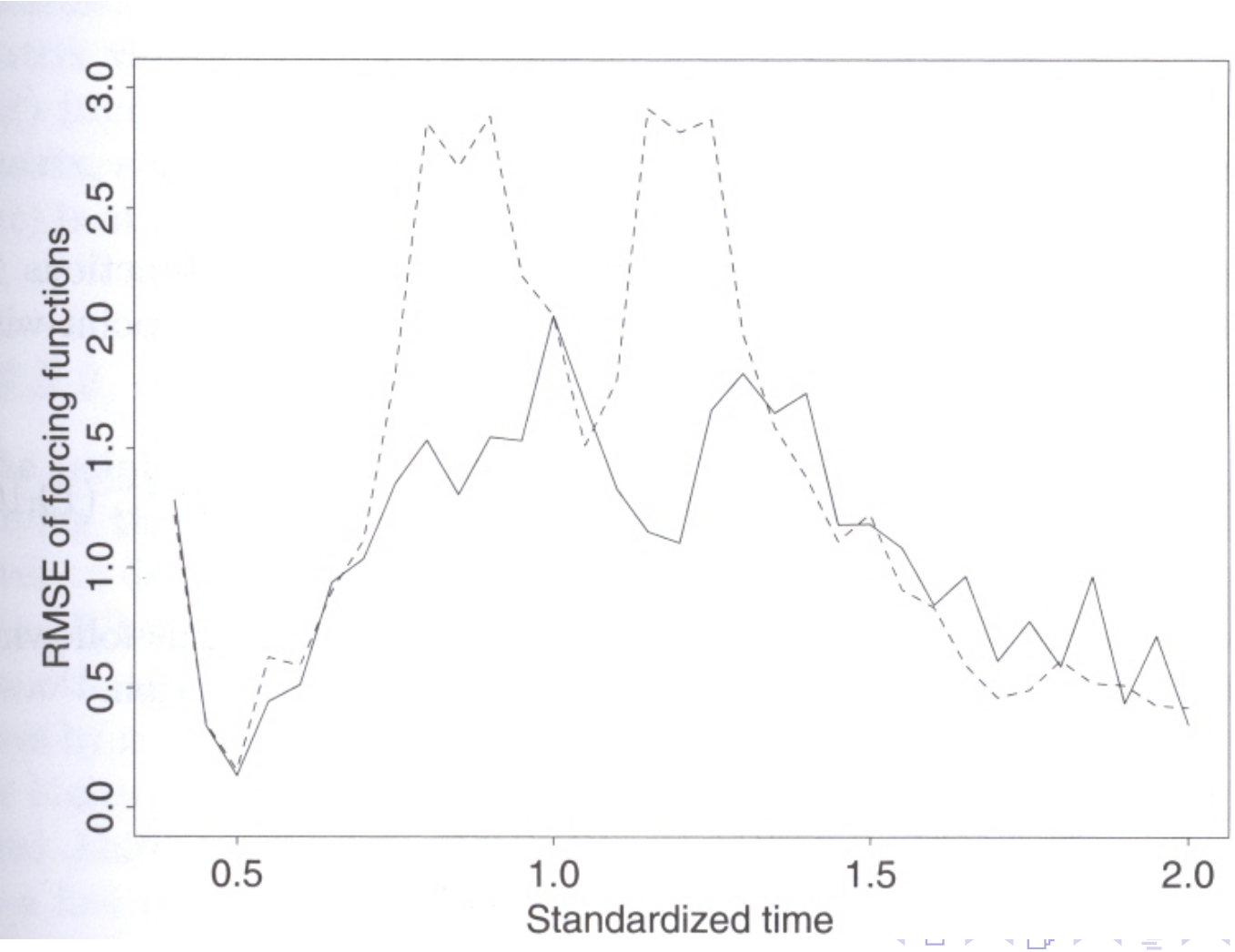


Forcing functions



Root MS of forcing functions

Solid: Estimated operator. Dashed: Theoretical operator.



Comparisons with the theoretical model

- The forcing function for the theoretical operator shows a systematic trend, while the empirical forcing function from PDA shows much less pattern.
- The estimated operator is superior to the theoretical operator in the epochs just before and after the peak force, where it produces a forcing function about half of the size of its theoretical counterpart.
- Conjecture: such an improvement from the estimated operator is probably due to drag or viscosity in the thumb-forefinger joint.

A point-wise approach

- The point-wise fitting criterion is

$$\begin{aligned}\mathrm{PSSE}_L(t) &= \sum_{i=1}^{N}[Lx_i(t) - f_i(t)]^2 \\ &= \sum_{i=1}^{N}[\sum_{j=0}^{m-1}\beta_j(t)D^j x_i(t) + D^m x_i(t) - f_i(t)]^2.\end{aligned}$$

- Define $\boldsymbol{\beta}(t) = (\beta_0(t), \ldots, \beta_{m-1}(t))'$, $\mathbf{y} = (D^m x_1(t) - f_1(t), \ldots, D^m x_N(t) - f_N(t))'$, and the $N \times m$ point-wise design matrix $\mathbf{Z}$ with rows

$$\mathbf{z}_i(t) = (-x_i(t), \ldots, -D^{m-1}x_i(t))'.$$

- Then the point-wise solution is

$$\hat{\boldsymbol{\beta}(t)} = [\mathbf{Z}(t)'\mathbf{Z}(t)]^{-1}\mathbf{Z}(t)'\mathbf{y}(t).$$

Drawbacks of the point-wise approach

- Solution may not be available on a fine resolution due to the smoothness constraints on higher derivatives, whose smoothness is generally not guaranteed by a smoothing procedure.
- Weight function from the point-wise approach may not be smooth.
- Can often run into singularity problem at a local area, while such problems can be circumvented by enforcing smoothness on the weight function.
- Requires $m < N$.

A concurrent model approach

- Define $\boldsymbol{\beta}(t)$, $\mathbf{y}$, and $\mathbf{Z}$ as before.
- The fitting criterion is the sum of **integrated** squared errors

$$\begin{aligned}\mathrm{ISSE}_L(t) &= \sum_{i=1}^{N} \int [Lx_i(t) - f_i(t)]^2 dt \\ &= \sum_{i=1}^{N} \int [y_i(t) - \mathbf{z}_i(t)'\boldsymbol{\beta}(t)]^2 dt,\end{aligned}$$

  which can also incorporating smoothness constraint by attaching a roughness penalty.
- Computational details same as concurrent models.

An iterative concurrent model approach

- Step 1: Use the concurrent model approach to obtain a smooth estimate of $\beta(t)$ with a chosen initial roughness penalty.
- Step 2: Use the estimate of $\beta(t)$ to form a linear differential operator L.
- Step 3: Refit the concurrent model with the roughness penalty defined by L.
- Iterate Steps 2-3 as needed.

An artificial example

A sample of 1000 sets of functional data were generated from the tilted sinusoidal model

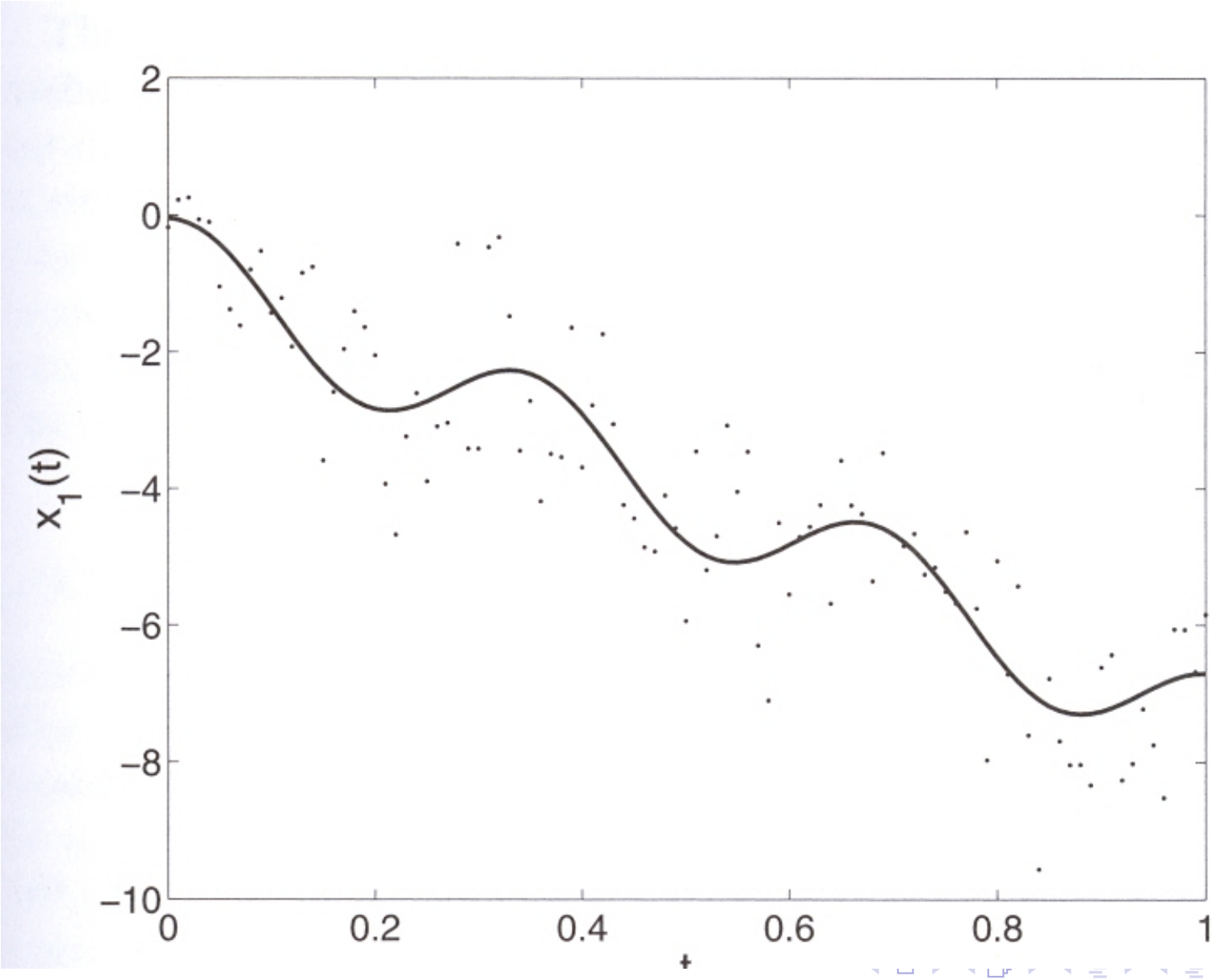
$$x_i(t_j) = c_{i1} + c_{i2}t_j + c_{i3}\sin(6\pi t_j) + c_{i4}\cos(6\pi t_j) + \epsilon_{ij}.$$

- time grid: $t_j = (0, 0.01, \ldots, 1)$.
- coefficients c_{ik} independently generated from a normal distribution with mean zero.
- Errors ϵ_{ij}: i.i.d. standard normal.
- The annihilating differential operator

$$L = (6\pi)^2 D^2 + D^4$$

where $(6\pi)^2 = 355.3$.

Simulation data



Initial setup

- Initial operator: $L = D^4$.
- Basis system: 105 order 6 B-spline basis functions with knots at the sampling points t_j.

Iterations

- Use a numerical optimization method to find the minimizer λ of the GCV criterion with the roughness penalty defined by the operator L. Enforce an upper limit $10 - \log_{10} \text{trace}(\mathbf{R})$ on the allowable estimate to avoid rounding error problems, where $\mathbf{R}$ is the penalty matrix.
- Smooth the data using the λ thus obtained.
- Perform a PDA based on the concurrent linear model method described before. Use the constant basis to estimate all the four coefficient functions $\beta_j, j = 0, 1, 2, 3$ (i.e., treat β_j's as constant functions).
- Update the linear differential operator L and redefine the penalty matrix $\mathbf{R}$.

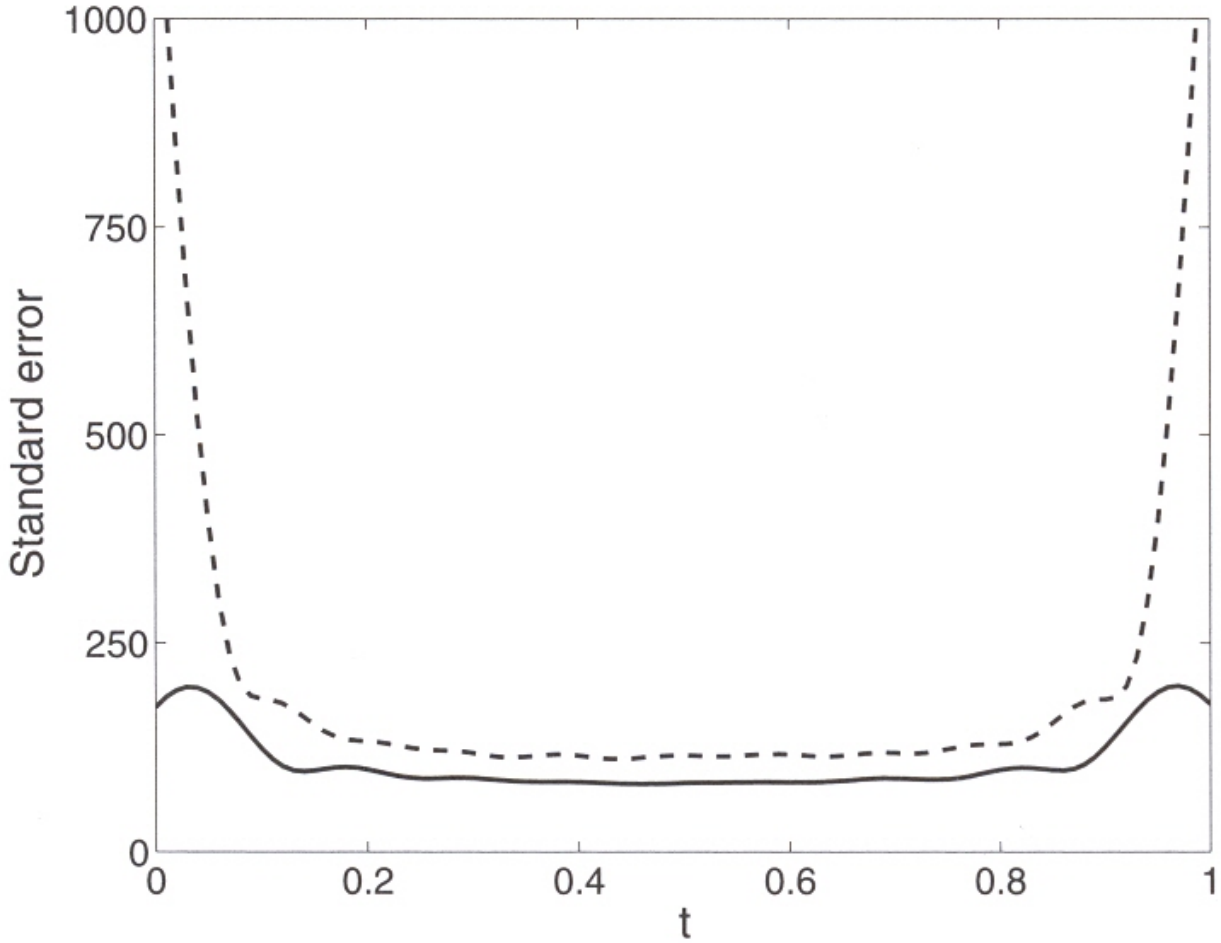
Results

- After the first iteration,
  - $\lambda = 10^{-9.9}$,
  - $L = 2360.3I - 123.8D + 376.1D^2 - 0.3D^3 + D^4$.
- After the second iteration,
  - $\lambda = 10^{-8.8}$, hardly change afterwards
- After the fifth iteration,
  - $L = 310.4I - 125.4D + 357.0D^2 - 0.4D^3 + D^4$.

Improvements with iterations

- The ratios of the first iteration IMSE to that on the fifth iteration are 1.2, 2.0, 3.8, 5.3, and 8.1 for derivatives of order 0, 1, 2, 3 and 4.
- A modest improvement on the function values but more improvements for higher order derivatives.
- Shown by the next plot, the biggest impact is at the endpoints, where estimating the linkage between the function value and the second derivatives greatly diminishes the standard error.

Std. Err. of the 2nd derivative after the 1st and 5th iterations



Assessing fit in PDA

- Define the ESS of a theoretical model and its associated weight functions ω_j

$$\mathrm{PSSE}_0(t) = \sum_{i=1}^{N} \left[\{ \sum_{j=0}^{m-1} \omega_j(t)(D^j x_i)(t) + D^m x_i(t) \} - f_i(t) \right]^2.$$

- When there is no theoretical model, use $\omega_j = 0$.
- Define the point-wise R-square:

$$\mathrm{RSQ}(t) = (\mathrm{PSSE}_0(t) - \mathrm{PSSE}_L(t))/\mathrm{PSSE}_0(t).$$

- Define the point-wise F-ratio:

$$\mathrm{FRATIO}(t) = \frac{(\mathrm{PSSE}_0(t) - \mathrm{PSSE}_L(t))/m}{\mathrm{PSSE}_0(t)/(N-m)}.$$

Principal Differential Analysis

- PDA fits a differential equation to noisy data to capture the features of a single curve or the variation in features across curves.
- It produces estimated differential equations that describe the underlying "mechanics" in the system of interest.
- It can be a good complement even if there is a theoretical model available.
- The iterative fitting procedure allows the refinement of the differential operator estimate as well as the smoothed function estimates.
- Can be generalized into the fitting of systems of equations, called *dynamic models*.